

In my initial post, I explained how, by achieving a balance between automation and human-centered design, resilience, and sustainability, Industry 5.0 can be used to resolve some problems in such industries as aviation. I included the 2024 CrowdStrike IT outage (especially Delta Air Lines) as an example (Abdel-Baset et al., 2025). The case concerns the unsuccessful update of security software, which caused extensive system failures, canceling thousands of flights and disrupting over 1.3 million passengers. I highlighted that principles of Industry 5.0, i.e., decentralization, redundancy of the systems, and human-in-the-loop protocols, could have alleviated the crisis. This resonates with the necessity of healthy IT risk control within industries such as aviation, with its central values of customer trust and business continuity (Alojaiman, 2023).

Ali Alhammadi replied, attributing the cascading failures of centralized systems to the ability to cause serious reputational harm and derail related sectors such as hospitality and tourism. He pointed out that Industry 5.0 implies a human-centered approach and resilience, critical to such failure deterrence. He also proposed adding more elaborate resilience solutions, such as backup control rooms, anomaly detection, and refined crisis communications, which would have limited the degree of the disruption.

Abdulla Almessabi further accentuated that centrally based IT systems are prone to crashing, which impacts operations and customers' confidence. He agreed with diversified infrastructure and human control (Moenck et al., 2025). He also cited the need for credible ethics between technology vendors and consumers in developing resilience and trust within the aviation industry (Alzaabi, 2024). He suggested that the technological infrastructure ought to be upgraded with the principles of Industry 5.0 in mind to prevent such interferences.

In summary, the argument returns to the point that all the technological changes that may serve in Industry 5.0 are great opportunities, but they should be used with care and vision, thus providing resiliency, security, and customer confidence even during unscheduled downtimes.

References

Abdel-Baset, M., Mohamed, R., & Chang, V. (2025). A multi-criteria decision-making framework to evaluate the impact of Industry 5.0 technologies: case study, lessons learned, challenges, and future directions. *Information Systems Frontiers*, 27(2), 791-821. Available at: <https://doi.org/10.1007/s10796-024-10472-3> (Accessed 13 August 2025).

Alojaiman, B. (2023). Technological modernizations in the Industry 5.0 era: A descriptive analysis and future research directions. *Processes*, 11(5), 1318. Available at: <https://doi.org/10.3390/pr11051318> (Accessed 13 August 2025).

Alzaabi, A. S. (2024). Optimizing safety in the age of Industry 5.0: Mitigating the domino effect. Available at: <https://khazna.ku.ac.ae/ws/portalfiles/portal/27091826/file> (Accessed 13 August 2025).

Moenck, K., Koch, J., Rath, J. E., Büsch, L., Giercker, J., Kähler, F., ... & Schüppstuhl, T. (2025). Industry 5.0 in aircraft production and MRO: challenges and opportunities. *CEAS Aeronautical Journal*, 1-25. Available at: <https://doi.org/10.1007/s13272-025-00832-3> (Accessed 13 August 2025).